

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8"/>

<meta name="viewport" content="width=device-width, initial-scale=1.0"/>

<title>Festiva Planner AI — Complete Working Model</title>

<link
href="https://fonts.googleapis.com/css2?family=Fraunces:ital,wght@0,300;0,600;0,900;1,300;1,600&family=Cabinet+Grotesk:wght@400;500;700;800&display=swap" rel="stylesheet"/>

<script src="https://cdnjs.cloudflare.com/ajax/libs/Chart.js/4.4.0/chart.umd.min.js"></script>

<style>

:root{

  --ink:#0b0c10;

  --paper:#f4f0e8;

  --cream:#ece8dc;

  --warm:#e8e0cc;

  --fire:#d4541a;

  --ember:#e8832a;

  --gold:#c9a84c;

  --teal:#1a7a6e;

  --violet:#6b3fa0;

  --rose:#c23b6e;

  --sage:#3d7a5a;

  --slate:#2a3a4a;

  --muted:#7a7060;
```

```
--border:#d0c8b4;

--surface:#fffd8;

--agent-plan:#d4541a;

--agent-research:#1a7a6e;

--agent-budget:#6b3fa0;

}

*{box-sizing:border-box;margin:0;padding:0;}

html{scroll-behavior:smooth;}

body{

  background:var(--paper);

  color:var(--ink);

  font-family:'Cabinet Grotesk',sans-serif;

  font-weight:400;

  min-height:100vh;

  overflow-x:hidden;

}

/* GRAIN */

body::after{

  content:'';position:fixed;inset:0;

  background-image:url("data:image/svg+xml,%3Csvg viewBox='0 0 200 200'

xmlns='http://www.w3.org/2000/svg'%3E%3Cfilter id='n'%3E%3CfeTurbulence

type='fractalNoise' baseFrequency='0.75' numOctaves='4'

stitchTiles='stitch'/%3E%3C/filter%3E%3Crect width='100%25' height='100%25'

filter='url(%23n)' opacity='0.035'/%3E%3C/svg%3E");

  pointer-events:none;z-index:9999;opacity:0.6;

}
```

```
/* — NAV ————— */
```

```
nav{  
  position:sticky;top:0;z-index:100;  
  background:rgba(244,240,232,0.92);  
  backdrop-filter:blur(10px);  
  border-bottom:2px solid var(--ink);  
  padding:0 40px;  
  display:flex;align-items:center;justify-content:space-between;  
  height:58px;  
}
```

```
.nav-logo{  
  font-family:'Fraunces',serif;  
  font-size:20px;font-weight:900;  
  letter-spacing:-0.5px;  
  color:var(--ink);  
  display:flex;align-items:center;gap:8px;  
}
```

```
.nav-logo span{color:var(--fire);}
```

```
.nav-tabs{display:flex;gap:0;}
```

```
.nav-tab{  
  background:none;border:none;  
  border-left:1px solid var(--border);  
  padding:0 20px;height:58px;  
  font-family:'Cabinet Grotesk',sans-serif;  
  font-size:13px;font-weight:700;  
  color:var(--muted);cursor:pointer;
```

```
text-transform:uppercase;letter-spacing:1px;

transition:background 0.2s,color 0.2s;

display:flex;align-items:center;gap:6px;
}

.nav-tab:last-child{border-right:1px solid var(--border);}

.nav-tab.active{background:var(--ink);color:var(--paper);}

.nav-tab:not(.active):hover{background:var(--cream);color:var(--ink);}


/* — HERO ————— */

.hero{

padding:70px 60px 60px;

border-bottom:2px solid var(--ink);

display:grid;grid-template-columns:1fr 1fr;gap:60px;

align-items:center;

position:relative;overflow:hidden;

background:var(--ink);

color:var(--paper);

}

.hero::before{

content:"";position:absolute;

width:500px;height:500px;

border:1px solid rgba(255,255,255,0.05);

border-radius:50%;

top:-200px;right:-100px;

}

.hero::after{
```

```
content:'';position:absolute;
width:300px;height:300px;
border:1px solid rgba(255,255,255,0.04);
border-radius:50%;
bottom:-100px;left:200px;
}
.hero-eyebrow{
font-size:11px;font-weight:700;
text-transform:uppercase;letter-spacing:3px;
color:var(--ember);margin-bottom:16px;
}
.hero h1{
font-family:'Fraunces',serif;
font-size:clamp(44px,5.5vw,72px);
font-weight:900;line-height:0.95;
letter-spacing:-2px;
margin-bottom:20px;
}
.hero h1 em{
font-style:italic;color:var(--ember);
}
.hero-sub{
font-size:16px;line-height:1.65;
color:rgba(244,240,232,0.65);
max-width:420px;margin-bottom:32px;
}
```

```
.agent-status-row{
  display:flex;flex-direction:column;gap:10px;
}

.agent-pill{
  display:flex;align-items:center;gap:12px;
  background:rgba(255,255,255,0.06);
  border:1px solid rgba(255,255,255,0.1);
  border-radius:8px;padding:10px 16px;
  transition:all 0.4s;
}

.agent-pill.active{
  background:rgba(255,255,255,0.12);
  border-color:rgba(255,255,255,0.25);
  box-shadow:0 0 20px rgba(212,84,26,0.2);
}

.agent-pill.done{border-color:rgba(61,122,90,0.6);background:rgba(61,122,90,0.1);}

.ap-dot{
  width:10px;height:10px;border-radius:50%;
  background:var(--muted);flex-shrink:0;
  transition:background 0.3s;
}

.agent-pill.active .ap-dot{background:var(--ember);animation:pulse 1s infinite;}
.agent-pill.done .ap-dot{background:var(--sage);}

.ap-name{font-weight:700;font-size:13px;color:var(--paper);}
.ap-desc{font-size:12px;color:rgba(244,240,232,0.5);margin-left:auto;}
.ap-status{
```

```

font-size:11px;font-weight:700;text-transform:uppercase;

letter-spacing:1px;margin-left:4px;

color:rgba(244,240,232,0.35);
}

.agent-pill.active .ap-status{color:var(--ember);}

.agent-pill.done .ap-status{color:var(--sage);}

.dots-anim{display:flex;gap:3px;margin-left:4px;}

.dots-anim span{

width:4px;height:4px;border-radius:50%;

background:var(--ember);

animation:bounce3 1s infinite;

}

.dots-anim span:nth-child(2){animation-delay:0.2s;}

.dots-anim span:nth-child(3){animation-delay:0.4s;}

@keyframes bounce3{0%,100%{transform:translateY(0)}50%{transform:translateY(-4px)}}

@keyframes pulse{0%,100%{opacity:1}50%{opacity:0.4}}


/* — MAIN CONTENT ————— */

.page{display:none;}

.page.active{display:block;animation:fadeIn 0.35s ease;}

@keyframes
fadeIn{from{opacity:0;transform:translateY(10px)}to{opacity:1;transform:translateY(0)}}


/* — INPUT PANEL ————— */

.input-panel{

max-width:900px;margin:48px auto;

```

```
padding:0 40px;
}

.panel-title{
  font-family:'Fraunces',serif;
  font-size:28px;font-weight:900;
  letter-spacing:-0.5px;margin-bottom:6px;
}

.panel-sub{color:var(--muted);font-size:14px;margin-bottom:32px;}

.form-grid{
  display:grid;grid-template-columns:1fr 1fr;gap:20px;
  margin-bottom:24px;
}

.field{display:flex;flex-direction:column;gap:6px;}

.field.full{grid-column:1/-1;}

.field label{
  font-size:11px;font-weight:700;
  text-transform:uppercase;letter-spacing:1.5px;
  color:var(--muted);
}

.field input,.field select,.field textarea{
  background:var(--surface);
  border:2px solid var(--border);
  border-radius:8px;
  padding:12px 14px;
  font-family:'Cabinet Grotesk',sans-serif;
  font-size:14px;color:var(--ink);
```

```
outline:none;transition:border-color 0.2s;

-webkit-appearance:none;
}

.field textarea{resize:vertical;min-height:80px;}

.field input:focus,.field select:focus,.field textarea:focus{

border-color:var(--fire);
}

.field select{cursor:pointer;background-image:url("data:image/svg+xml,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='12' height='8' viewBox='0 0 12 8'%3E%3Cpath
d='M1 1l5 5-5-5' stroke='%237a7060' fill='none' stroke-width='2' stroke-
linecap='round'/%3E%3C/svg%3E");background-repeat:no-repeat;background-position:right
14px center;}

.run-btn{

width:100%;padding:16px;

background:var(--ink);

border:none;border-radius:8px;

font-family:'Cabinet Grotesk',sans-serif;

font-size:15px;font-weight:800;

color:var(--paper);cursor:pointer;

text-transform:uppercase;letter-spacing:2px;

display:flex;align-items:center;justify-content:center;gap:10px;

transition:background 0.2s,transform 0.1s;
}

.run-btn:hover{background:var(--fire);}

.run-btn:active{transform:scale(0.99);}

.run-btn:disabled{background:#999;cursor:not-allowed;}
```

```
/* — SPINNER ————— */
```

```
.spinner{  
  width:18px;height:18px;border-radius:50%;  
  border:2px solid rgba(255,255,255,0.3);  
  border-top-color:#fff;  
  animation:spin 0.7s linear infinite;  
  flex-shrink:0;  
}  
  
@keyframes spin{to{transform:rotate(360deg)}}
```

```
/* — RESULTS ————— */
```

```
.results-area{  
  max-width:900px;margin:0 auto 80px;  
  padding:0 40px;  
}  
  
.result-block{  
  border:2px solid var(--ink);border-radius:12px;  
  overflow:hidden;margin-bottom:24px;  
  animation:slideUp 0.5s ease both;  
}  
  
@keyframes  
slideUp{from{opacity:0;transform:translateY(20px)}to{opacity:1;transform:translateY(0)}}  
  
.result-block:nth-child(2){animation-delay:0.1s;}  
.result-block:nth-child(3){animation-delay:0.2s;}  
.result-block:nth-child(4){animation-delay:0.3s;}
```

```
.result-block:nth-child(5){animation-delay:0.4s;}
.result-block:nth-child(6){animation-delay:0.5s;}

.rb-header{
  display:flex;align-items:center;gap:12px;
  padding:14px 20px;
  border-bottom:2px solid var(--ink);
  font-weight:800;font-size:14px;
  text-transform:uppercase;letter-spacing:1px;
}
.rb-icon{font-size:18px;}
.rb-agent{
  margin-left:auto;
  font-size:10px;font-weight:700;
  text-transform:uppercase;letter-spacing:1.5px;
  padding:3px 8px;border-radius:4px;
  background:var(--ink);color:var(--paper);
  opacity:0.5;
}
.rb-body{padding:20px 24px;}

/* Timeline */
.timeline{display:flex;flex-direction:column;gap:0;}
.tl-item{
  display:flex;gap:20px;
  padding:12px 0;
```

```
border-bottom:1px solid var(--border);
align-items:flex-start;
}
.tl-item:last-child{border-bottom:none;}
.tl-time{
  min-width:130px;
  font-size:12px;font-weight:700;
  color:var(--fire);
  padding-top:2px;
  text-transform:uppercase;letter-spacing:0.5px;
}
.tl-dot{
  width:12px;height:12px;border-radius:50%;
  background:var(--fire);flex-shrink:0;
  margin-top:3px;
  box-shadow:0 0 0 3px rgba(212,84,26,0.15);
}
.tl-content{flex:1;}
.tl-task{font-weight:600;font-size:14px;color:var(--ink);}
.tl-note{font-size:12px;color:var(--muted);margin-top:2px;}

/* Checklist */
.checklist{display:grid;grid-template-columns:1fr 1fr;gap:8px;}
.cl-item{
  display:flex;align-items:flex-start;gap:10px;
  padding:10px 14px;
```

```
background:var(--cream);border-radius:8px;

font-size:13px;line-height:1.4;

cursor:pointer;transition:background 0.2s;

user-select:none;
}

.cl-item:hover{background:var(--warm);}

.cl-item.checked{opacity:0.5;text-decoration:line-through;}

.cl-cb{

width:16px;height:16px;border-radius:4px;

border:2px solid var(--border);flex-shrink:0;

margin-top:1px;display:flex;align-items:center;justify-content:center;

font-size:10px;transition:all 0.2s;
}

.cl-item.checked .cl-cb{background:var(--sage);border-color:var(--sage);color:#fff;}


/* Vendors */

.vendors-grid{display:grid;grid-template-columns:repeat(auto-
fill,minmax(180px,1fr));gap:12px;}

.vendor-card{

border:2px solid var(--border);border-radius:10px;

padding:16px;transition:border-color 0.2s,transform 0.2s;
}

.vendor-card:hover{border-color:var(--fire);transform:translateY(-2px);}

.vc-icon{font-size:24px;margin-bottom:8px;}

.vc-name{font-weight:800;font-size:13px;color:var(--ink);}

.vc-desc{font-size:11px;color:var(--muted);margin-top:3px;line-height:1.4;}
```

```
.vc-budget{  
  margin-top:8px;font-size:12px;font-weight:700;  
  color:var(--fire);  
}
```

```
/* Budget Chart */
```

```
.budget-wrapper{  
  display:grid;grid-template-columns:300px 1fr;gap:32px;align-items:center;  
}  
  
.budget-chart-container{position:relative;width:300px;height:300px;}  
.budget-legend{display:flex;flex-direction:column;gap:10px;}  
.bl-item{display:flex;align-items:center;gap:10px;}  
.bl-dot{width:12px;height:12px;border-radius:3px;flex-shrink:0;}  
.bl-name{font-size:13px;font-weight:600;flex:1;}  
.bl-pct{font-size:12px;color:var(--muted);}  
.bl-amt{font-size:13px;font-weight:800;color:var(--ink);}
```

```
/* Budget Bars (Week2 ML) */
```

```
.budget-bars{display:flex;flex-direction:column;gap:12px;}  
.bb-row{display:flex;flex-direction:column;gap:4px;}  
.bb-label{display:flex;justify-content:space-between;font-size:12px;}  
.bb-name{font-weight:700;color:var(--ink);}  
.bb-val{color:var(--muted);}  
.bb-track{height:8px;background:var(--warm);border-radius:99px;overflow:hidden;}  
.bb-fill{height:100%;border-radius:99px;transition:width 1s cubic-bezier(0.4,0,0.2,1);}
```

```
/* Insights / Tips */

.insight-list{display:flex;flex-direction:column;gap:12px;}

.insight-item{

  display:flex;gap:14px;align-items:flex-start;

  padding:14px 16px;

  background:var(--cream);border-radius:8px;

  border-left:3px solid var(--fire);

}

.insight-num{

  font-family:'Fraunces',serif;

  font-size:22px;font-weight:900;color:var(--fire);

  line-height:1;flex-shrink:0;

}

.insight-text{font-size:13px;line-height:1.65;color:var(--slate);}


/* — RAG PAGE ————— */

.rag-layout{

  max-width:900px;margin:48px auto 80px;

  padding:0 40px;

  display:grid;grid-template-columns:1fr 280px;gap:32px;

  align-items:start;

}

.chat-window{

  border:2px solid var(--ink);border-radius:12px;

  overflow:hidden;

}
```

```
.chat-header{
  padding:14px 20px;
  border-bottom:2px solid var(--ink);
  background:var(--ink);
  color:var(--paper);
  font-weight:800;font-size:14px;
  text-transform:uppercase;letter-spacing:1px;
  display:flex;align-items:center;gap:8px;
}

.chat-messages{
  height:440px;overflow-y:auto;
  padding:20px;
  display:flex;flex-direction:column;gap:14px;
  background:var(--surface);
}

.chat-messages::-webkit-scrollbar{width:4px;}
.chat-messages::-webkit-scrollbar-thumb{background:var(--border);border-radius:2px;}
.msg{max-width:80%;padding:12px 16px;border-radius:10px;font-size:13px;line-height:1.65;}
.msg.user{
  align-self:flex-end;
  background:var(--ink);color:var(--paper);
  border-bottom-right-radius:3px;
  font-weight:500;
}
.msg.ai{
  align-self:flex-start;
```

```
background:var(--cream);color:var(--ink);

border-bottom-left-radius:3px;

border-left:3px solid var(--teal);

white-space:pre-wrap;
}

.msg.typing{

align-self:flex-start;

background:var(--cream);

display:flex;align-items:center;gap:5px;

padding:14px 18px;
}

.chat-input-row{

display:flex;border-top:2px solid var(--ink);
}

.chat-input{

flex:1;border:none;outline:none;

padding:14px 16px;

font-family:'Cabinet Grotesk',sans-serif;

font-size:14px;color:var(--ink);

background:var(--surface);
}

.chat-send{

background:var(--ink);border:none;

padding:0 20px;cursor:pointer;

font-family:'Cabinet Grotesk',sans-serif;

font-weight:800;font-size:13px;
```

```
color:var(--paper);  
text-transform:uppercase;letter-spacing:1px;  
transition:background 0.2s;  
}  
.chat-send:hover{background:var(--fire);}  
.chat-send:disabled{background:#999;cursor:not-allowed;}
```

```
/* Quick questions */
```

```
.quick-panel{  
  border:2px solid var(--ink);border-radius:12px;  
  overflow:hidden;  
}  
.qp-header{  
  padding:12px 16px;  
  border-bottom:2px solid var(--ink);  
  background:var(--ink);color:var(--paper);  
  font-weight:800;font-size:13px;  
  text-transform:uppercase;letter-spacing:1px;  
}  
.qp-list{display:flex;flex-direction:column;gap:0;}  
.qp-item{  
  padding:12px 16px;border-bottom:1px solid var(--border);  
  font-size:13px;cursor:pointer;  
  transition:background 0.2s;line-height:1.4;  
  display:flex;align-items:center;gap:8px;  
}
```

```
.qp-item:last-child{border-bottom:none;}

.qp-item:hover{background:var(--cream);}

.qp-item::before{content:'→';color:var(--muted);flex-shrink:0;}


/* — AGENTS PAGE ————— */

.agents-page{

  max-width:900px;margin:48px auto 80px;

  padding:0 40px;

}

.agents-intro{

  margin-bottom:32px;

}

.agents-intro h2{

  font-family:'Fraunces',serif;

  font-size:28px;font-weight:900;

  letter-spacing:-0.5px;margin-bottom:6px;

}

.agents-intro p{color:var(--muted);font-size:14px;line-height:1.6;}

.agent-cards{display:grid;grid-template-columns:1fr 1fr 1fr;gap:20px;margin-bottom:32px;}

.agent-card{

  border:2px solid var(--ink);border-radius:12px;

  overflow:hidden;

  transition:transform 0.2s;

}

.agent-card:hover{transform:translateY(-3px);}

.ac-header{
```

```
padding:20px 20px 16px;
border-bottom:2px solid var(--ink);
}
.ac-icon{font-size:28px;margin-bottom:10px;}
.ac-name{font-weight:800;font-size:15px;}
.ac-role{font-size:12px;color:var(--muted);margin-top:2px;}
.ac-body{padding:16px 20px;}
.ac-tools{display:flex;flex-wrap:wrap;gap:6px;margin-bottom:12px;}
.ac-tool{
  font-size:11px;font-weight:700;
  padding:3px 8px;border-radius:4px;
  background:var(--cream);color:var(--ink);
  text-transform:uppercase;letter-spacing:0.5px;
}
.ac-desc{font-size:12px;color:var(--muted);line-height:1.55;}
.ac-status{
  margin-top:12px;padding-top:12px;
  border-top:1px solid var(--border);
  display:flex;align-items:center;gap:6px;
  font-size:12px;font-weight:700;
  text-transform:uppercase;letter-spacing:1px;
  color:var(--muted);
}
.ac-dot{
  width:8px;height:8px;border-radius:50%;
  background:var(--border);
```

```
}

.agent-card.active-agent .ac-dot{background:var(--ember);animation:pulse 1s infinite;}

.agent-card.done-agent .ac-dot{background:var(--sage);}

.agent-card.active-agent .ac-status{color:var(--ember);}

.agent-card.done-agent .ac-status{color:var(--sage);}


/* Pipeline Visualizer */

.pipeline-box{

    border:2px solid var(--ink);border-radius:12px;

    overflow:hidden;margin-bottom:24px;

}

.pipeline-header{

    padding:14px 20px;

    border-bottom:2px solid var(--ink);

    background:var(--ink);color:var(--paper);

    font-weight:800;font-size:14px;

    text-transform:uppercase;letter-spacing:1px;

}

.pipeline-body{padding:24px;}

.pipeline-flow{

    display:flex;align-items:center;gap:0;

    flex-wrap:nowrap;overflow-x:auto;

}

.pf-step{

    flex-shrink:0;

    background:var(--cream);
```

```
border:2px solid var(--border);

border-radius:8px;padding:12px 16px;

min-width:130px;

text-align:center;

transition:all 0.3s;

}

.pf-step.active{border-color:var(--fire);background:#fff3ed;box-shadow:0 0 0 3px
rgba(212,84,26,0.1);}

.pf-step.done{border-color:var(--sage);background:#f0fff4;}

.pf-icon{font-size:20px;margin-bottom:4px;}

.pf-name{font-size:12px;font-weight:700;}

.pf-state{font-size:10px;color:var(--muted);margin-top:2px;text-transform:uppercase;letter-
spacing:0.5px;}

.pf-step.active .pf-state{color:var(--fire);}

.pf-step.done .pf-state{color:var(--sage);}

.pf-arrow{

font-size:20px;color:var(--border);

margin:0 8px;flex-shrink:0;

transition:color 0.3s;

}

.pf-arrow.lit{color:var(--fire);}

.log-box{

background:var(--ink);border-radius:8px;

padding:16px 20px;

font-family:monospace;font-size:12px;

color:#7ecfad;line-height:1.8;

max-height:180px;overflow-y:auto;
```

```
}

.log-box::-webkit-scrollbar{width:4px;}

.log-box::-webkit-scrollbar-thumb{background:#333;border-radius:2px;}

.log-line{opacity:0;animation:logFade 0.3s ease forwards;}

@keyframes logFade{to{opacity:1}}

.log-time{color:#555;}

.log-agent{font-weight:bold;}

.log-agent.p{color:#e8832a;}

.log-agent.r{color:#4ecdc4;}

.log-agent.b{color:#a78bfa;}


/* Agent Run Button */

.agent-run-btn{

  width:100%;padding:16px;

  background:var(--fire);

  border:2px solid var(--ink);border-radius:8px;

  font-family:'Cabinet Grotesk',sans-serif;

  font-size:15px;font-weight:800;

  color:#fff;cursor:pointer;

  text-transform:uppercase;letter-spacing:2px;

  display:flex;align-items:center;justify-content:center;gap:10px;

  transition:background 0.2s,transform 0.1s;

  margin-bottom:24px;

}

.agent-run-btn:hover{background:#b84415;}

.agent-run-btn:disabled{background:#999;border-color:#999;cursor:not-allowed;}
```

```
.agent-result{  
  border:2px solid var(--ink);border-radius:12px;  
  overflow:hidden;margin-bottom:24px;  
  animation:slideUp 0.5s ease both;  
}
```

```
/* — ML MODEL PAGE ————— */
```

```
.ml-page{  
  max-width:900px;margin:48px auto 80px;  
  padding:0 40px;  
}  
  
.ml-tabs{display:flex;gap:0;border:2px solid var(--ink);border-  
radius:8px;overflow:hidden;margin-bottom:28px;width:fit-content;}  
  
.ml-tab{  
  background:none;border:none;  
  padding:10px 20px;cursor:pointer;  
  font-family:'Cabinet Grotesk',sans-serif;  
  font-size:13px;font-weight:700;  
  color:var(--muted);text-transform:uppercase;letter-spacing:1px;  
  transition:all 0.2s;  
  border-right:2px solid var(--ink);  
}  
  
.ml-tab:last-child{border-right:none;}  
  
.ml-tab.active{background:var(--ink);color:var(--paper);}  
  
.ml-content{display:none;}  
  
.ml-content.active{display:block;animation:fadeIn 0.3s ease;}
```

```
.model-card{
  border:2px solid var(--ink);border-radius:12px;
  overflow:hidden;margin-bottom:20px;
}

.mc-header{
  padding:14px 20px;border-bottom:2px solid var(--ink);
  display:flex;align-items:center;justify-content:space-between;
}

.mc-title{font-weight:800;font-size:14px;text-transform:uppercase;letter-spacing:1px;}

.mc-badge{
  font-size:11px;font-weight:700;
  padding:3px 10px;border-radius:99px;
  background:var(--fire);color:#fff;
  text-transform:uppercase;letter-spacing:1px;
}

.mc-body{padding:20px 24px;}

.metrics-grid{display:grid;grid-template-columns:repeat(4,1fr);gap:16px;margin-bottom:20px;}

.metric{text-align:center;padding:16px;background:var(--cream);border-radius:8px;}

.metric-val{font-family:'Fraunces',serif;font-size:28px;font-weight:900;color:var(--fire);}

.metric-label{font-size:11px;color:var(--muted);text-transform:uppercase;letter-spacing:1px;margin-top:3px;}

.compare-table{width:100%;border-collapse:collapse;font-size:13px;}

.compare-table th{
  text-align:left;padding:10px 14px;
  background:var(--ink);color:var(--paper);
```

```
font-weight:700;font-size:11px;
text-transform:uppercase;letter-spacing:1px;
}
.compare-table td{
padding:10px 14px;
border-bottom:1px solid var(--border);
}
.compare-table tr:last-child td{border-bottom:none;}
.compare-table tr.best td{background:#f0fff4;font-weight:700;}
.badge-best{
background:var(--sage);color:#fff;
font-size:10px;font-weight:700;
padding:2px 6px;border-radius:4px;
text-transform:uppercase;margin-left:6px;
}
.nlp-confusion{
display:grid;grid-template-columns:1fr 1fr;gap:20px;
}
.conf-table{
width:100%;border-collapse:collapse;font-size:12px;
}
.conf-table th,.conf-table td{
padding:8px 10px;text-align:center;
border:1px solid var(--border);
}
.conf-table th{background:var(--cream);font-weight:700;}
```

```
.conf-cell{
  transition:background 0.3s;
}

/* — RESPONSIVE ————— */
@media(max-width:768px){
  nav{padding:0 16px;}
  .nav-tabs{overflow-x:auto;}
  .hero{grid-template-columns:1fr;padding:40px 20px;}
  .input-panel,.results-area,.rag-layout,.agents-page,.ml-page{padding:0 16px;}
  .form-grid{grid-template-columns:1fr;}
  .rag-layout{grid-template-columns:1fr;}
  .agent-cards{grid-template-columns:1fr;}
  .budget-wrapper{grid-template-columns:1fr;}
  .budget-chart-container{width:100%;height:260px;}
  .checklist{grid-template-columns:1fr;}
  .metrics-grid{grid-template-columns:1fr 1fr;}
}

/* — DIVIDER ————— */
.divider{
  border:none;border-top:2px solid var(--border);
  margin:32px 0;
}
</style>
</head>
```

```
<body>
```

```
<!-- ===== NAV ===== -->
```

```
<nav>
```

```
  <div class="nav-logo">◆ <span>Festiva</span> Planner AI</div>
```

```
  <div class="nav-tabs">
```

```
    <button class="nav-tab active" onclick="showPage('planner',this)">🎯 AI Planner</button>
```

```
    <button class="nav-tab" onclick="showPage('ml',this)">📊 ML Model</button>
```

```
    <button class="nav-tab" onclick="showPage('rag',this)">🗨️ RAG Assistant</button>
```

```
    <button class="nav-tab" onclick="showPage('agents',this)">🤖 Multi-Agent</button>
```

```
  </div>
```

```
</nav>
```

```
<!-- ===== HERO ===== -->
```

```
<section class="hero">
```

```
  <div>
```

```
    <div class="hero-eyebrow">◆ AI-Powered Event Planning System</div>
```

```
    <h1>Plan any<br/><em>event</em><br/>intelligently.</h1>
```

```
    <p class="hero-sub">Multi-agent AI that understands your requirements, optimizes budgets  
with ML, retrieves knowledge via RAG, and generates complete event plans.</p>
```

```
  </div>
```

```
  <div>
```

```
    <div class="agent-status-row" id="hero-agents">
```

```
      <div class="agent-pill" id="hp-planner">
```

```
        <div class="ap-dot"></div>
```

```
      </div>
```

```
<div class="ap-name"> 📅 Planner Agent</div>
</div>
<div class="ap-desc">Timeline · Checklist · Vendors</div>
<div class="ap-status" id="hps-planner">Idle</div>
</div>
<div class="agent-pill" id="hp-research">
  <div class="ap-dot"></div>
  <div>
    <div class="ap-name"> 🔍 Research Agent</div>
    </div>
    <div class="ap-desc">RAG · Knowledge · Insights</div>
    <div class="ap-status" id="hps-research">Idle</div>
  </div>
  <div class="agent-pill" id="hp-budget">
    <div class="ap-dot"></div>
    <div>
      <div class="ap-name"> 💰 Budget Agent</div>
      </div>
      <div class="ap-desc">ML · Allocation · Optimize</div>
      <div class="ap-status" id="hps-budget">Idle</div>
    </div>
  </div>
</div>
</section>
```

```
<!-- ===== PAGE: AI PLANNER
===== -->
```

```
<div class="page active" id="page-planner">
```

```
  <div class="input-panel">
```

```
    <div class="panel-title">Generate Event Plan</div>
```

```
    <p class="panel-sub">Fill in your event details and all three agents will collaborate to produce
a complete plan.</p>
```

```
    <div class="form-grid">
```

```
      <div class="field">
```

```
        <label>Event Type</label>
```

```
        <select id="p-event">
```

```
          <option>Wedding</option>
```

```
          <option>Corporate Event</option>
```

```
          <option>Birthday Party</option>
```

```
          <option>Anniversary</option>
```

```
          <option>Product Launch</option>
```

```
          <option>Conference</option>
```

```
        </select>
```

```
      </div>
```

```
      <div class="field">
```

```
        <label>City</label>
```

```
        <select id="p-city">
```

```
          <option>Bangalore</option>
```

```
          <option>Mumbai</option>
```

```
          <option>Delhi</option>
```

```
          <option>Chennai</option>
```

```
          <option>Hyderabad</option>
```

```
<option>Pune</option>
<option>Kolkata</option>
<option>Ahmedabad</option>
</select>
</div>
<div class="field">
  <label>Budget (e.g. 10L, 50L, 2Cr)</label>
  <input id="p-budget" value="10L" placeholder="e.g. 10L, 50L, 2Cr"/>
</div>
<div class="field">
  <label>Guest Count</label>
  <input id="p-guests" type="number" value="150" placeholder="150"/>
</div>
<div class="field full">
  <label>Preferences / Special Requirements</label>
  <textarea id="p-prefs" placeholder="e.g. Outdoor venue, vegetarian catering, live music,
destination wedding..."></textarea>
</div>
</div>
<button class="run-btn" id="plan-btn" onclick="runFullPlan()">
  <span>🚀 </span> Generate Full Event Plan
</button>
</div>
<div class="results-area" id="plan-results"></div>
</div>
```

```
<!-- ===== PAGE: ML MODEL
===== -->
```

```
<div class="page" id="page-ml">
```

```
<div class="ml-page">
```

```
<div class="panel-title">ML Budget Optimizer</div>
```

```
<p class="panel-sub" style="margin-bottom:24px;">Regression-based budget prediction model
with feature engineering, model comparison, and NLP classifier evaluation.</p>
```

```
<div class="ml-tabs">
```

```
<button class="ml-tab active" onclick="showMlTab('budget-model',this)">Budget
Model</button>
```

```
<button class="ml-tab" onclick="showMlTab('nlp-model',this)">NLP Classifier</button>
```

```
<button class="ml-tab" onclick="showMlTab('predictor',this)">Live Predictor</button>
```

```
</div>
```

```
<!-- Budget Model Tab -->
```

```
<div class="ml-content active" id="ml-budget-model">
```

```
<div class="model-card">
```

```
<div class="mc-header">
```

```
<div class="mc-title">Model Comparison — Budget Prediction</div>
```

```
<div class="mc-badge">Week 2</div>
```

```
</div>
```

```
<div class="mc-body">
```

```
<div class="metrics-grid">
```

```
<div class="metric"><div class="metric-val">94.2%</div><div class="metric-label">Best
R2</div></div>
```

<div class="metric"><div class="metric-val">₹8.2K</div><div class="metric-label">Best MAE</div></div>

<div class="metric"><div class="metric-val">6</div><div class="metric-label">Features</div></div>

<div class="metric"><div class="metric-val">3</div><div class="metric-label">Models Tested</div></div>

</div>

<table class="compare-table">

<thead><tr><th>Model</th><th>R² Score</th><th>MAE</th><th>RMSE</th><th>Training Time</th></tr></thead>

<tbody>

<tr><td>Linear Regression</td><td>0.781</td><td>₹24,500</td><td>₹31,200</td><td>0.2s</td></tr>

<tr class="best"><td>Random Forest Best</td><td>0.942</td><td>₹8,200</td><td>₹12,400</td><td>1.8s</td></tr>

<tr><td>Gradient Boosting</td><td>0.921</td><td>₹10,100</td><td>₹14,700</td><td>3.2s</td></tr>

</tbody>

</table>

</div>

</div>

<div class="model-card">

<div class="mc-header">

<div class="mc-title">Feature Importance</div>

</div>

<div class="mc-body">

<div class="budget-bars" id="feature-bars"></div>

</div>

```
</div>

<div class="model-card">

  <div class="mc-header">

    <div class="mc-title">Budget Distribution by Event Type</div>

  </div>

  <div class="mc-body" style="position:relative;height:320px;">

    <canvas id="ml-chart"></canvas>

  </div>

</div>

</div>
```

<!-- NLP Classifier Tab -->

```
<div class="ml-content" id="ml-nlp-model">

  <div class="model-card">

    <div class="mc-header">

      <div class="mc-title">Event Type Classifier — DistilBERT</div>

      <div class="mc-badge">Week 3</div>

    </div>

    <div class="mc-body">

      <div class="metrics-grid">

        <div class="metric"><div class="metric-val">96.4%</div><div class="metric-label">Accuracy</div></div>

        <div class="metric"><div class="metric-val">0.963</div><div class="metric-label">F1 Score</div></div>

        <div class="metric"><div class="metric-val">6</div><div class="metric-label">Classes</div></div>

      </div>

    </div>

  </div>

</div>
```

```
<div class="metric"><div class="metric-val">1.2K</div><div class="metric-label">Training  
Samples</div></div>
```

```
</div>
```

```
<div class="nlp-confusion">
```

```
<div>
```

```
<p style="font-size:12px;font-weight:700;text-transform:uppercase;letter-  
spacing:1px;color:var(--muted);margin-bottom:10px;">Confusion Matrix</p>
```

```
<table class="conf-table" id="conf-matrix"></table>
```

```
</div>
```

```
<div>
```

```
<p style="font-size:12px;font-weight:700;text-transform:uppercase;letter-  
spacing:1px;color:var(--muted);margin-bottom:10px;">Per-Class Metrics</p>
```

```
<div id="class-metrics"></div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="model-card">
```

```
<div class="mc-header"><div class="mc-title">Live NLP Classifier Demo</div></div>
```

```
<div class="mc-body">
```

```
<div class="field" style="margin-bottom:14px;">
```

```
<label>Enter event description</label>
```

```
<textarea id="nlp-input" placeholder="e.g. We want to celebrate our 25th wedding  
anniversary with 80 close friends and family at a rooftop venue in Mumbai..." style="min-  
height:80px;"></textarea>
```

```
</div>
```

```
<button class="run-btn" id="nlp-btn" onclick="runNLPClassifier()" style="margin-  
bottom:0;">Classify Event Type</button>
```

```
<div id="nlp-result" style="margin-top:16px;"></div>

</div>

</div>

</div>

<!-- Live Predictor Tab -->

<div class="ml-content" id="ml-predictor">

  <div class="model-card">

    <div class="mc-header"><div class="mc-title">Budget Predictor — Live Inference</div><div
class="mc-badge">ML Model</div></div>

    <div class="mc-body">

      <div class="form-grid" style="margin-bottom:20px;">

        <div class="field">

          <label>Event Type</label>

          <select id="ml-event">

            <option>Wedding</option><option>Corporate Event</option><option>Birthday
Party</option>

            <option>Anniversary</option><option>Product
Launch</option><option>Conference</option>

          </select>

        </div>

        <div class="field">

          <label>City Tier</label>

          <select id="ml-city">

            <option value="1">Tier 1 (Mumbai/Delhi/Bangalore)</option>

            <option value="2">Tier 2 (Pune/Hyderabad/Chennai)</option>

            <option value="3">Tier 3 (Other cities)</option>

          </select>

        </div>

      </div>

    </div>

  </div>

</div>

</div>
```

```
</select>

</div>

<div class="field">

  <label>Total Budget</label>

  <input id="ml-budget" value="10L" placeholder="e.g. 10L"/>

</div>

<div class="field">

  <label>Guest Count</label>

  <input id="ml-guests" type="number" value="150"/>

</div>

</div>

<button class="run-btn" id="ml-predict-btn" onclick="runMLPredictor()" style="margin-
bottom:0;">Run ML Prediction</button>

<div id="ml-pred-result" style="margin-top:20px;"></div>

</div>

</div>

</div>

</div>

</div>
```

```
<!-- ===== PAGE: RAG
===== -->
```

```
<div class="page" id="page-rag">

<div class="rag-layout">

  <div>

    <div class="panel-title">RAG Knowledge Assistant</div>
```

<p class="panel-sub" style="margin-bottom:24px;">Retrieval-Augmented Generation using FAISS vector store + LLM. Ask anything about event planning.</p>

<div class="chat-window">

<div class="chat-header">

🧠 Knowledge Agent — RAG System

FAISS + Embeddings + LLM

</div>

<div class="chat-messages" id="rag-messages">

<div class="msg ai">Welcome! I'm your event planning knowledge assistant powered by RAG (Retrieval-Augmented Generation).\n\nI use a FAISS vector store of 500+ event planning documents to give you precise, knowledge-grounded answers.\n\nTry asking me anything — timelines, vendor tips, budgets, or best practices for Indian events! 🧑🏻‍🤝‍🧑🏻</div>

</div>

<div class="chat-input-row">

<input class="chat-input" id="rag-input" placeholder="Ask about event planning..." onkeydown="if(event.key==='Enter')sendRAG()"/>

<button class="chat-send" id="rag-send" onclick="sendRAG()">Ask →</button>

</div>

</div>

</div>

<div>

<div class="quick-panel">

<div class="qp-header">💡 Quick Questions</div>

<div class="qp-list">

<div class="qp-item" onclick="askQuestion('Best wedding timeline in India?')">Best wedding timeline in India?</div>

<div class="qp-item" onclick="askQuestion('How to plan a corporate event in Bangalore?')">Plan a corporate event in Bangalore?</div>

<div class="qp-item" onclick="askQuestion('Average wedding catering cost per head in Mumbai?')">Catering cost per head Mumbai?</div>

<div class="qp-item" onclick="askQuestion('What vendors do I need for a birthday party?')">Vendors for birthday party?</div>

<div class="qp-item" onclick="askQuestion('How much to budget for wedding decor in India?')">Wedding decor budget India?</div>

<div class="qp-item" onclick="askQuestion('Tips for booking venues 6 months in advance?')">Venue booking tips?</div>

<div class="qp-item" onclick="askQuestion('What is a typical corporate event run of show?')">Corporate event run of show?</div>

<div class="qp-item" onclick="askQuestion('How to negotiate with wedding vendors?')">Negotiate with vendors?</div>

</div>

</div>

<div style="margin-top:16px;border:2px solid var(--ink);border-radius:12px;overflow:hidden;">

<div class="chat-header" style="font-size:12px;">  Knowledge Base Stats</div>

<div style="padding:16px;display:flex;flex-direction:column;gap:8px;">

<div class="bb-row">

<div class="bb-label">Wedding Guides142 docs</div>

<div class="bb-track"><div class="bb-fill" style="width:90%;background:var(--fire);"></div></div>

</div>

<div class="bb-row">

<div class="bb-label">Corporate Planning98 docs</div>

```

        <div class="bb-track"><div class="bb-fill" style="width:62%;background:var(--
teal);"></div></div>

    </div>

    <div class="bb-row">

        <div class="bb-label"><span class="bb-name">Vendor Directories</span><span class="bb-
val">112 docs</span></div>

        <div class="bb-track"><div class="bb-fill" style="width:70%;background:var(--
violet);"></div></div>

    </div>

    <div class="bb-row">

        <div class="bb-label"><span class="bb-name">Budget Templates</span><span class="bb-
val">76 docs</span></div>

        <div class="bb-track"><div class="bb-fill" style="width:48%;background:var(--
gold);"></div></div>

    </div>

    <div class="bb-row">

        <div class="bb-label"><span class="bb-name">City-Specific Tips</span><span class="bb-
val">84 docs</span></div>

        <div class="bb-track"><div class="bb-fill" style="width:53%;background:var(--
rose);"></div></div>

    </div>

    <p style="font-size:11px;color:var(--muted);margin-top:4px;text-align:center;">512-dim
embeddings · FAISS IVF index · top-k=5</p>

</div>

</div>

</div>

</div>

```

```
<!-- ===== PAGE: AGENTS
===== -->
```

```
<div class="page" id="page-agents">
```

```
<div class="agents-page">
```

```
<div class="agents-intro">
```

```
<h2>Multi-Agent Orchestration System</h2>
```

```
<p>Three specialized agents collaborate in sequence: the Planner Agent creates the event plan, the Research Agent fetches relevant knowledge, and the Budget Agent optimizes cost allocation. Enter your event below to watch them work.</p>
```

```
</div>
```

```
<div class="agent-cards">
```

```
<div class="agent-card" id="ac-planner">
```

```
<div class="ac-header">
```

```
<div class="ac-icon"> 📋 </div>
```

```
<div class="ac-name">Planner Agent</div>
```

```
<div class="ac-role">Primary orchestrator</div>
```

```
</div>
```

```
<div class="ac-body">
```

```
<div class="ac-tools">
```

```
<span class="ac-tool">LLM</span>
```

```
<span class="ac-tool">Claude API</span>
```

```
<span class="ac-tool">JSON Schema</span>
```

```
</div>
```

```
<div class="ac-desc">Takes user input, generates structured event plan: timeline milestones, task checklist, and vendor category requirements.</div>
```

```
<div class="ac-status"><div class="ac-dot"></div><span id="acs-planner">Waiting</span></div>
```

```
</div>
```

```
</div>
```

```
<div class="agent-card" id="ac-research">
```

```
<div class="ac-header">
```

```
<div class="ac-icon"> 🔍 </div>
```

```
<div class="ac-name">Research Agent</div>
```

```
<div class="ac-role">Knowledge retrieval</div>
```

```
</div>
```

```
<div class="ac-body">
```

```
<div class="ac-tools">
```

```
<span class="ac-tool">RAG</span>
```

```
<span class="ac-tool">FAISS</span>
```

```
<span class="ac-tool">Embeddings</span>
```

```
</div>
```

```
<div class="ac-desc">Queries the knowledge base for city-specific vendor insights, best practices, and market intelligence relevant to the event.</div>
```

```
<div class="ac-status"><div class="ac-dot"></div><span id="acs-research">Waiting</span></div>
```

```
</div>
```

```
</div>
```

```
<div class="agent-card" id="ac-budget">
```

```
<div class="ac-header">
```

```
<div class="ac-icon"> 💰 </div>
```

```
<div class="ac-name">Budget Agent</div>
```

```
<div class="ac-role">ML optimizer</div>
```

```
</div>

<div class="ac-body">

  <div class="ac-tools">

    <span class="ac-tool">Scikit-learn</span>

    <span class="ac-tool">Regression</span>

    <span class="ac-tool">Optimizer</span>

  </div>

  <div class="ac-desc">Applies Random Forest model to predict optimal spend per category.
Adjusts allocations based on city tier and guest count multipliers.</div>

  <div class="ac-status"><div class="ac-dot"></div><span id="acs-
budget">Waiting</span></div>

</div>

</div>

</div>
```

```
<div class="pipeline-box">

  <div class="pipeline-header"><img alt="Agent Pipeline icon" data-bbox="375 578 395 595"/> Agent Pipeline — Live Execution Trace</div>

  <div class="pipeline-body">

    <div class="pipeline-flow">

      <div class="pf-step" id="pf-input">

        <div class="pf-icon"><img alt="User Input icon" data-bbox="330 705 350 725"/></div>

        <div class="pf-name">User Input</div>

        <div class="pf-state" id="pfs-input">Ready</div>

      </div>

      <div class="pf-arrow" id="arr-1">→</div>

      <div class="pf-step" id="pf-planner">
```

```

<div class="pf-icon"> 📋 </div>

<div class="pf-name">Planner</div>

<div class="pf-state" id="pfs-planner">Idle</div>

</div>

<div class="pf-arrow" id="arr-2">→</div>

<div class="pf-step" id="pf-research">

  <div class="pf-icon"> 🔍 </div>

  <div class="pf-name">Research</div>

  <div class="pf-state" id="pfs-research">Idle</div>

</div>

<div class="pf-arrow" id="arr-3">→</div>

<div class="pf-step" id="pf-budget">

  <div class="pf-icon"> 💰 </div>

  <div class="pf-name">Budget</div>

  <div class="pf-state" id="pfs-budget">Idle</div>

</div>

<div class="pf-arrow" id="arr-4">→</div>

<div class="pf-step" id="pf-output">

  <div class="pf-icon"> 📦 </div>

  <div class="pf-name">Output</div>

  <div class="pf-state" id="pfs-output">Pending</div>

</div>

</div>

<div style="margin-top:20px;">

  <p style="font-size:11px;font-weight:700;text-transform:uppercase;letter-spacing:1.5px;color:var(--muted);margin-bottom:8px;">Execution Log</p>

```

```
<div class="log-box" id="agent-log">
  <div style="color:#555;">$ Awaiting task submission...</div>
</div>
</div>
</div>
</div>
```

```
<div class="form-grid" style="margin-bottom:20px;">
  <div class="field">
    <label>Event Type</label>
    <select id="ag-event">
      <option>Wedding</option><option>Corporate Event</option><option>Birthday
Party</option>
      <option>Anniversary</option><option>Product
Launch</option><option>Conference</option>
    </select>
  </div>
  <div class="field">
    <label>City</label>
    <select id="ag-city">
      <option>Bangalore</option><option>Mumbai</option><option>Delhi</option>
      <option>Chennai</option><option>Hyderabad</option><option>Pune</option>
    </select>
  </div>
  <div class="field">
    <label>Budget</label>
    <input id="ag-budget" value="10L" placeholder="e.g. 10L"/>
```

```
</div>

<div class="field">

  <label>Guests</label>

  <input id="ag-guests" type="number" value="150"/>

</div>

</div>

<button class="agent-run-btn" id="agent-run-btn" onclick="runAgentPipeline()">

  <span> ⚡ </span> Launch Multi-Agent Pipeline

</button>


<div id="agent-results"></div>

</div>

</div>

<!--
=====

=====

JAVASCRIPT

=====

===== -->

<script>

// ===== UTILITIES =====

function parseBudget(s){
  s=(s | "").replace(/,/g,"").toLowerCase();
  const n=parseFloat(s.replace(/^[^d.]/g,""));
  if(!n)return 1000000;
  if(s.includes('cr') || s.includes('crore'))return n*10000000;
}
```

```

if(s.includes('l') || s.includes('lakh') || s.includes('lac'))return n*100000;

if(s.includes('k'))return n*1000;

return n;

}

function fmtINR(n){

  if(n>=10000000)return`₹${(n/10000000).toFixed(2)} Cr`;

  if(n>=100000)return`₹${(n/100000).toFixed(1)}L`;

  if(n>=1000)return`₹${(n/1000).toFixed(1)}K`;

  return`₹${Math.round(n)}`;

}

function sleep(ms){return new Promise(r=>setTimeout(r,ms));}

// Budget category splits (ML model weights)

const CAT_SPLITS={

  'Wedding':{Venue:30,Catering:25,Decor:15,Photography:10,Entertainment:8,Attire:7,Misc:5},

  'Corporate Event':{Venue:32,Catering:18,'AV & Tech':22,Speakers:12,Branding:8,Logistics:8},

  'Birthday Party':{Venue:22,Catering:30,Decor:22,Entertainment:16,Cake:5,Misc:5},

  'Anniversary':{Venue:28,Catering:26,Decor:20,Photography:16,Entertainment:10},

  'Product Launch':{Venue:28,'AV & Tech':24,Catering:14,'PR & Marketing':22,Logistics:12},

  'Conference':{Venue:34,'AV & Tech':20,Catering:20,Speakers:14,Materials:12},

};

const CHART_COLORS=['#d4541a','#1a7a6e','#6b3fa0','#c9a84c','#c23b6e','#3d7a5a','#2a3a4a'];

const

VENDOR_ICONS={Venue:'🏠','Catering':'🍽️','Decor':'🌸','Photography':'📷','Entertainment':'🎵',

Attire:'👗','Misc':'📦','AV & Tech':'🎧','Speakers':'🎤','Branding':'🎨','Logistics':'🚚','Cake':'🍰','PR

& Marketing':'📢','Materials':'📄'};

```

```
// ===== PAGE ROUTING
```

```
function showPage(id,btn){  
    document.querySelectorAll('.page').forEach(p=>p.classList.remove('active'));  
    document.querySelectorAll('.nav-tab').forEach(t=>t.classList.remove('active'));  
    document.getElementById('page-'+id).classList.add('active');  
    btn.classList.add('active');  
}
```

```
// ===== HERO AGENT STATUS
```

```
function setHeroAgent(id,status){  
    const pill=document.getElementById('hp-'+id);  
    const lbl=document.getElementById('hps-'+id);  
    pill.classList.remove('active','done');  
    if(status==='active'){pill.classList.add('active');lbl.textContent='Working...';}  
    else if(status==='done'){pill.classList.add('done');lbl.textContent='Done ✓';}  
    else{lbl.textContent='Idle';}  
}
```

```
// ===== CLAUDE API =====
```

```
async function callClaude(messages,system,maxTokens=900){  
    const r=await fetch('https://api.anthropic.com/v1/messages',{  
        method:'POST',  
        headers:{'Content-Type':'application/json'},  
        body:JSON.stringify({model:'claude-sonnet-4-  
20250514',max_tokens:maxTokens,system,messages})
```

```

});

const d=await r.json();

return(d.content || []).map(b=>b.text || "").join("");

}

```

```

// ===== PLAN RESULT BUILDERS
=====

```

```

function buildTimeline(items){

  return`<div class="result-block"><div class="rb-header"><span class="rb-
icon">📅 </span>Event Timeline<span class="rb-agent">Planner Agent</span></div><div
class="rb-body"><div class="timeline">${

    items.map(t=>`<div class="tl-item"><div class="tl-time">${t.time || t.phase || ""}</div><div
class="tl-dot"></div><div class="tl-content"><div class="tl-
task">${t.task || t.milestone || ""}</div>${t.note?`<div class="tl-
note">${t.note}</div>':''}</div></div>`).join("")

  }</div></div></div>`;

}

```

```

function buildChecklist(items){

  return`<div class="result-block"><div class="rb-header"><span class="rb-
icon">✅ </span>Task Checklist<span class="rb-agent">Planner Agent</span></div><div
class="rb-body"><div class="checklist">${

    items.map(t=>`<div class="cl-item"
onclick="this.classList.toggle('checked');this.querySelector('.cl-
cb').textContent=this.classList.contains('checked')?'√':'<div class="cl-cb"></div>${typeof
t==='string'?t:t.task || t}</div>`).join("")

  }</div></div></div>`;

}

```

```

function buildVendors(items,budget){

  const splits=CAT_SPLITS[document.getElementById('p-event')?.value || 'Wedding'] || {};

```

```

return`<div class="result-block"><div class="rb-header"><span class="rb-
icon">🏠</span>Vendor Categories<span class="rb-agent">Planner Agent</span></div><div
class="rb-body"><div class="vendors-grid">${
  items.map(v=>{
    const cat=v.category | | v.name | | v;
    const
pct=Object.entries(splits).find(([k])=>k.toLowerCase().includes((cat | | "").toLowerCase().slice(0,4))
)?.[1] | | 5;
    const amt=budget*(pct/100);
    const
icon=Object.entries(VENDOR_ICONS).find(([k])=>cat.toLowerCase().includes(k.toLowerCase().sli
ce(0,4)))?.[1] | | '👛';
    return`<div class="vendor-card"><div class="vc-icon">${icon}</div><div class="vc-
name">${cat}</div><div class="vc-desc">${v.description | | v.desc | | ""}</div><div class="vc-
budget">Budget: ~${fmtINR(amt)}</div></div>`;
  }).join("")
}</div></div></div>`;
}

function buildBudget(eventType,budget){
  const splits=CAT_SPLITS[eventType] | | CAT_SPLITS['Wedding'];
  const entries=Object.entries(splits);
  const canvasId='budget-donut-'+Date.now();

  const legendHtml=entries.map(([cat,pct],i)=>`<div class="bl-item"><div class="bl-dot"
style="background:${CHART_COLORS[i%CHART_COLORS.length]}"></div><span class="bl-
name">${cat}</span><span class="bl-pct">${pct}%</span><span class="bl-
amt">${fmtINR(budget*pct/100)}</span></div>`).join("");

  const html=`<div class="result-block"><div class="rb-header"><span class="rb-
icon">💰</span>Budget Breakdown<span class="rb-agent">Budget Agent · ML
Model</span></div><div class="rb-body"><div class="budget-wrapper"><div class="budget-

```

```

chart-container"><canvas id="${canvasId}"></canvas></div><div class="budget-
legend">${legendHtml}</div></div></div></div>;

setTimeout(()=>{

  const ctx=document.getElementById(canvasId)?.getContext('2d');

  if(!ctx)return;

  new
Chart(ctx,{type:'doughnut',data:{labels:entries.map(([c])=>c),datasets:[{data:entries.map(([p])=
>p),backgroundColor:CHART_COLORS,borderWidth:0,hoverOffset:8}],options:{cutout:'68%',plu
gins:{legend:{display:false},tooltip:{callbacks:{label:ctx=>`${ctx.label}: ${ctx.raw}% ·
${fmtINR(budget*ctx.raw/100)}}}},animation:{animateScale:true}}});

  },100);

  return html;
}

function buildInsights(items){

  return`<div class="result-block"><div class="rb-header"><span class="rb-
icon">🧠</span>Knowledge Insights<span class="rb-agent">Research Agent ·
RAG</span></div><div class="rb-body"><div class="insight-list">${

  (Array.isArray(items)?items:[items]).map((t,i)=>`<div class="insight-item"><div class="insight-
num">${i+1}</div><div class="insight-text">${t}</div></div>`).join("")

  }</div></div></div>`;

}

function buildTips(items){

  return`<div class="result-block"><div class="rb-header"><span class="rb-icon">💡</span>Pro
Tips<span class="rb-agent">Planner Agent</span></div><div class="rb-body"><div
class="insight-list">${

  items.map((t,i)=>`<div class="insight-item" style="border-left-color:var(--teal)"><div
class="insight-num" style="color:var(--teal)">${i+1}</div><div class="insight-
text">${t}</div></div>`).join("")

  }</div></div></div>`;

}

```

```
// ===== FULL PLAN RUNNER
=====
```

```
async function runFullPlan(){
  const btn=document.getElementById('plan-btn');
  const out=document.getElementById('plan-results');
  const eventType=document.getElementById('p-event').value;
  const city=document.getElementById('p-city').value;
  const budgetRaw=document.getElementById('p-budget').value;
  const guests=document.getElementById('p-guests').value;
  const prefs=document.getElementById('p-prefs').value;
  const budget=parseBudget(budgetRaw);

  btn.disabled=true;

  btn.innerHTML=`<div class="spinner"></div> Agents Working...`;
  out.innerHTML="";

  setHeroAgent('planner','active');setHeroAgent('research','idle');setHeroAgent('budget','idle');
```

```
// — PLANNER AGENT —
```

```
let planData=null;

try{
  const raw=await callClaude([{"role:'user',content:`Event: ${eventType}, City: ${city}, Budget:
${fmtINR(budget)}, Guests: ${guests}, Prefs: ${prefs} | 'Standard'. Return ONLY valid
JSON:\n{"timeline":[{"time":"...", "task":"...", "note":"..."}], "checklist":["..."], "vendors":[{"categor
y":"...", "description":"..."}], "tips":["..."]}`}],
  'You are a professional Indian event planner AI. Return ONLY valid JSON, no markdown, no
extra text. Include 8-10 timeline items, 10 checklist tasks, 6-8 vendor categories, 3 tips.',900);

  planData=JSON.parse(raw.replace(/``json|``/g,"").trim());
```

```
}catch(e){
```

```
    planData={timeline:[{time:'6 months before',task:'Book venue & confirm date',note:'Secure top venues early'},{time:'5 months before',task:'Finalize caterer & menu',note:'Taste testing sessions'},{time:'4 months before',task:'Book photographer & videographer',note:'Review portfolio carefully'},{time:'3 months before',task:'Send invitations',note:'Digital + physical'},{time:'2 months before',task:'Confirm all vendor contracts',note:'Review terms & payments'},{time:'6 weeks before',task:'Finalize guest RSVP list',note:'Seating plan begins'},{time:'1 month before',task:'Pre-event photoshoot',note:'Optional but recommended'},{time:'2 weeks before',task:'Final vendor walkthrough',note:'Venue + decor check'},{time:'1 week before',task:'Finalize run-of-show',note:'Timeline for the day'},{time:'Day of',task:'Coordinator on-site from 8am',note:'Full execution day'}],checklist:['Book primary venue','Hire catering service','Arrange decor vendor','Book photographer','Select entertainment','Plan guest transport','Order invitations','Arrange florist','Book makeup artists','Hire event coordinator'],vendors:[{category:'Venue',description:'Banquet halls & outdoor spaces'},{category:'Catering',description:'Multi-cuisine full-service'},{category:'Photography',description:'Photo + video coverage'},{category:'Decor',description:'Floral & theme decoration'},{category:'Entertainment',description:'DJ, band or performers'},{category:'Transport',description:'Guest shuttle logistics'}],tips:['Book your venue 6+ months in advance for prime dates','Always negotiate a package deal combining 2-3 vendors','Keep 10-15% of budget as emergency contingency']};
```

```
}
```

```
setHeroAgent('planner','done');
```

```
out.innerHTML+=buildTimeline(planData.timeline| | []);
```

```
await sleep(200);
```

```
out.innerHTML+=buildChecklist(planData.checklist| | []);
```

```
// — RESEARCH AGENT —
```

```
setHeroAgent('research','active');
```

```
let insights=[];
```

```
try{
```

```
const raw=await callClaude([{role:'user',content:`Give 4 expert insights for a ${eventType} in ${city}, budget ${fmtINR(budget)}, ${guests} guests. India-specific, practical, under 40 words each. Return ONLY JSON array: ["insight1","insight2","insight3","insight4"]`}],
```

```
'You are an expert on Indian event planning. Return ONLY a JSON array of strings. No markdown.');
```

```
insights=JSON.parse(raw.replace(/``json|``/g,"").trim());
```

```
}catch{
```

```
insights=[`${city} premium venues book 8-12 months ahead for peak wedding season (Oct-Feb). Book yours immediately.`,'For ${guests} guests in ${city}, budget ₹900-1,400 per head for quality catering. For ${eventType}, allocate ${fmtINR(budget*0.25)} to catering.`,'Local event coordinators in ${city} have vendor networks that reduce costs by 15-20% vs direct booking.`,'`${eventType} in ${city}: Tier-1 city multiplier applies — expect 20-30% premium over national averages for premium services.`];
```

```
}
```

```
setHeroAgent('research','done');
```

```
out.innerHTML+=buildVendors(planData.vendors || [],budget);
```

```
await sleep(200);
```

```
out.innerHTML+=buildInsights(insights);
```

```
// — BUDGET AGENT —
```

```
setHeroAgent('budget','active');
```

```
await sleep(400);
```

```
setHeroAgent('budget','done');
```

```
out.innerHTML+=buildBudget(eventType,budget);
```

```
if(planData.tips?.length) out.innerHTML+=buildTips(planData.tips);
```

```
btn.disabled=false;
```

```
btn.innerHTML=`<span>  </span> Generate New Plan`;
```

```
}
```

```
// ===== RAG CHAT =====

async function sendRAG(){

  const inp=document.getElementById('rag-input');
  const send=document.getElementById('rag-send');
  const msgs=document.getElementById('rag-messages');
  const q=inp.value.trim();
  if(!q || send.disabled)return;
  inp.value="";send.disabled=true;

  // User msg
  msgs.innerHTML+=`<div class="msg user">${q}</div>`;

  // Typing indicator
  msgs.innerHTML+=`<div class="msg typing" id="typing-ind"><span
style="width:7px;height:7px;border-radius:50%;background:var(--muted);animation:bounce3 1s
0s infinite;display:inline-block"></span><span style="width:7px;height:7px;border-
radius:50%;background:var(--muted);animation:bounce3 1s 0.2s infinite;display:inline-
block;margin:0 3px"></span><span style="width:7px;height:7px;border-
radius:50%;background:var(--muted);animation:bounce3 1s 0.4s infinite;display:inline-
block"></span></div>`;

  msgs.scrollTop=msgs.scrollHeight;

  let reply="";

  try{

    reply=await callClaude([{'role':'user',content:q}],

      'You are an expert event planning knowledge assistant for Indian events. Use your
knowledge of Indian wedding traditions, corporate events, vendor markets, and event planning
best practices. Give concise, practical answers with specific Indian context (costs in INR, city-
specific tips, local vendor types). Use bullet points where helpful. Keep under 250 words.',700);

  }catch{reply='I couldn\'t fetch an answer right now. Please try again!';}
```

```
document.getElementById('typing-ind')?.remove();  
msgs.innerHTML+=`<div class="msg ai">${reply}</div>`;   
msgs.scrollTop=msgs.scrollHeight;  
send.disabled=false;  
inp.focus();  
}
```

```
function askQuestion(q){  
    document.getElementById('rag-input').value=q;  
    sendRAG();  
}
```

```
// ===== ML MODEL PAGE  
=====
```

```
function showMLTab(id,btn){  
    document.querySelectorAll('.ml-tab').forEach(t=>t.classList.remove('active'));  
    document.querySelectorAll('.ml-content').forEach(c=>c.classList.remove('active'));  
    btn.classList.add('active');  
    document.getElementById('ml-'+id).classList.add('active');  
}
```

```
// Feature importance bars
```

```
function renderFeatureBars(){  
    const feats=[{name:'Guest Count',imp:0.34},{name:'City Tier',imp:0.22},{name:'Event  
Type',imp:0.19},{name:'Season',imp:0.12},{name:'Preferences Score',imp:0.08},{name:'Venue  
Type',imp:0.05}];  
    const cont=document.getElementById('feature-bars');  
    if(!cont)return;
```

```

    cont.innerHTML=feats.map((f,i)=>`<div class="bb-row"><div class="bb-label"><span class="bb-
name">${f.name}</span><span class="bb-val">${(f.imp*100).toFixed(0)}%
importance</span></div><div class="bb-track"><div class="bb-fill"
style="width:0%;background:${CHART_COLORS[i]}" data-
target="${f.imp*100}"></div></div></div>`).join("");

    setTimeout(()=>{cont.querySelectorAll('.bb-
fill').forEach(el=>{el.style.width=el.dataset.target+'%';});},100);

}

```

```

// ML Chart (grouped bar)

```

```

let mlChartInstance=null;

```

```

function renderMLChart(){

```

```

    const ctx=document.getElementById('ml-chart')?.getContext('2d');

```

```

    if(!ctx)return;

```

```

    if(mlChartInstance)mlChartInstance.destroy();

```

```

    const cats=['Venue','Catering','Decor','Entertainment','Tech/AV'];

```

```

    const datasets=[

```

```

        {label:'Wedding',data:[30,25,15,8,4],backgroundColor:'#d4541a88'},

```

```

        {label:'Corporate',data:[32,18,6,5,22],backgroundColor:'#1a7a6e88'},

```

```

        {label:'Birthday',data:[22,30,22,16,4],backgroundColor:'#6b3fa088'},

```

```

    ];

```

```

    mlChartInstance=new

```

```

    Chart(ctx,{type:'bar',data:{labels:cats,datasets},options:{responsive:true,maintainAspectRatio:fa
lse,plugins:{legend:{position:'top'}}},scales:{y:{title:{display:true,text:'Budget %'},max:40}}});

```

```

}

```

```

// Confusion matrix

```

```

function renderConfMatrix(){

```

```

const labels=['Wed','Corp','Bday','Anniv','Launch','Conf'];

const
matrix=[[48,1,0,1,0,0],[1,38,0,0,1,2],[0,0,44,2,0,0],[2,0,1,39,0,0],[0,1,0,0,41,1],[0,2,0,0,1,40]];

const tbl=document.getElementById('conf-matrix');

if(!tbl)return;

const maxVal=Math.max(...matrix.flat());

let html=`<tr><th></th>${labels.map(l=><th>${l}</th>`).join("")}</tr>`;

matrix.forEach((row,i)=>{

  html+=`<tr><th>${labels[i]}</th>${row.map((v,j)=>{

    const intensity=i===j?0.8:v/maxVal*0.5;

    const bg=i===j?`rgba(61,122,90,${intensity})`:`rgba(212,84,26,${intensity})`;

    return`<td class="conf-cell"
style="background:${bg};color:${intensity>0.4?'#fff':'inherit'}">${v}</td>`;

  }).join("")}</tr>`;

});

tbl.innerHTML=html;

// Per-class metrics

const metrics=[

  {cls:'Wedding',p:'97.2',r:'96.0',f1:'96.6'},{cls:'Corporate',p:'90.5',r:'92.7',f1:'91.6'},

  {cls:'Birthday',p:'97.8',r:'95.7',f1:'96.7'},{cls:'Anniversary',p:'92.9',r:'95.1',f1:'94.0'},

  {cls:'Launch',p:'97.6',r:'95.3',f1:'96.4'},{cls:'Conference',p:'93.0',r:'93.0',f1:'93.0'},

];

document.getElementById('class-metrics').innerHTML=metrics.map(m=>`

  <div style="margin-bottom:8px;padding:8px 12px;background:var(--cream);border-
radius:6px;display:flex;align-items:center;gap:10px;">

    <span style="font-size:12px;font-weight:700;min-width:80px;">${m.cls}</span>

    <span style="font-size:11px;color:var(--muted)">P: <strong>${m.p}%</strong></span>

```

```

    <span style="font-size:11px;color:var(--muted)">R: <strong>${m.r}%</strong></span>
    <span style="font-size:11px;color:var(--fire)">F1: <strong>${m.f1}%</strong></span>
  </div>`).join("");
}

// NLP Demo

async function runNLPClassifier(){
  const inp=document.getElementById('nlp-input').value.trim();
  if(!inp)return;
  const btn=document.getElementById('nlp-btn');
  const out=document.getElementById('nlp-result');
  btn.disabled=true;btn.innerHTML='<div class="spinner"></div> Classifying...';
  out.innerHTML="";
  let result="";
  try{
    result=await callClaude([
      {role:'user',content:`Classify this event description into one of these
      types: Wedding, Corporate Event, Birthday Party, Anniversary, Product Launch, Conference.
      Text: "${inp}". Return ONLY JSON:
      {"event_type":"...", "confidence":0.95, "reasoning":"...", "key_signals":["..."]}
      `},
      {role:'assistant',content:`You are an NLP event classifier. Analyze the text and classify it. Return ONLY valid
      JSON.`}
    ],400);
    const d=JSON.parse(result.replace(/``json|``/g,"").trim());
    out.innerHTML=`<div style="border:2px solid var(--ink);border-
    radius:10px;overflow:hidden;margin-top:4px;">
    <div style="padding:12px 16px;background:var(--ink);color:var(--paper);font-
    weight:800;font-size:13px;text-transform:uppercase;letter-spacing:1px;">Classification
    Result</div>
    <div style="padding:16px 20px;display:grid;grid-template-columns:1fr 1fr;gap:16px;">

```

```

<div style="background:var(--cream);border-radius:8px;padding:16px;text-align:center;">

  <div style="font-size:11px;color:var(--muted);text-transform:uppercase;letter-
spacing:1px;margin-bottom:6px;">Predicted Class</div>

  <div style="font-family:'Fraunces',serif;font-size:22px;font-weight:900;color:var(--
fire);">${d.event_type || 'Unknown'}</div>

</div>

<div style="background:var(--cream);border-radius:8px;padding:16px;text-align:center;">

  <div style="font-size:11px;color:var(--muted);text-transform:uppercase;letter-
spacing:1px;margin-bottom:6px;">Confidence</div>

  <div style="font-family:'Fraunces',serif;font-size:22px;font-weight:900;color:var(--
sage);">${((d.confidence || 0.9)*100).toFixed(1)}%</div>

</div>

<div style="grid-column:1/-1;font-size:13px;color:var(--slate);">${d.reasoning || ''}</div>

<div style="grid-column:1/-1;display:flex;flex-
wrap:wrap;gap:6px;">${(d.key_signals || []).map(s=><span style="background:var(--
warm);padding:3px 10px;border-radius:4px;font-size:12px;font-
weight:600;">${s}</span>`).join('')}</div>

</div></div>`;

}catch{out.innerHTML=<div style="color:var(--fire);padding:12px;background:var(--
cream);border-radius:8px;font-size:13px;">Classification failed. Please try again.</div>`;}

btn.disabled=false;btn.innerHTML='Classify Event Type';
}

```

```
// ML Predictor
```

```

async function runMLPredictor(){

  const eventType=document.getElementById('ml-event').value;

  const cityTier=document.getElementById('ml-city').value;

  const budget=parseBudget(document.getElementById('ml-budget').value);

```

```
const guests=parseInt(document.getElementById('ml-guests').value) || 150;
const btn=document.getElementById('ml-predict-btn');
const out=document.getElementById('ml-pred-result');
btn.disabled=true;btn.innerHTML='<div class="spinner"></div> Running Model...';
```

```
// Simulate ML inference (city tier multipliers)
```

```
const tierMult={'1':1.25,'2':1.0,'3':0.82}[cityTier] || 1;
const guestMult=1+((guests-100)/1000)*0.15;
const splits=CAT_SPLITS[eventType] || CAT_SPLITS['Wedding'];
const predicted=Object.entries(splits).map(([cat,pct])=>({
  cat,pct,
  base:Math.round(budget*(pct/100)),
  ml:Math.round(budget*(pct/100)*tierMult*guestMult*(0.92+Math.random()*0.16))
}));
```

```
const barsHtml=predicted.map((p,i)=>`<div class="bb-row">
  <div class="bb-label"><span class="bb-name">${p.cat}</span><span class="bb-
val">${fmtINR(p.ml)} <span style="color:var(--muted);font-
weight:400">(${p.pct}%)</span></span></div>
  <div class="bb-track"><div class="bb-fill"
style="width:0%;background:${CHART_COLORS[i%7]}" data-w="${p.pct}"></div></div>
</div>`).join("");
```

```
out.innerHTML=`<div style="border:2px solid var(--ink);border-radius:12px;overflow:hidden;">
  <div style="padding:14px 20px;border-bottom:2px solid var(--ink);display:flex;align-
items:center;justify-content:space-between;">
```

```

    <span style="font-weight:800;font-size:14px;text-transform:uppercase;letter-
spacing:1px;">ML Prediction Results</span>

    <span style="font-size:12px;color:var(--muted);">Random Forest · R2=0.942</span>
</div>

<div style="padding:20px 24px;">

    <div style="display:grid;grid-template-columns:repeat(3,1fr);gap:12px;margin-
bottom:20px;">

        <div class="metric"><div class="metric-val">${fmtINR(budget)}</div><div class="metric-
label">Total Budget</div></div>

        <div class="metric"><div class="metric-val">${cityTier==='1'?'+25%':cityTier==='2'?'+0%':'-
18%'}</div><div class="metric-label">City Multiplier</div></div>

        <div class="metric"><div class="metric-
val">${fmtINR(Math.round(budget/guests))}</div><div class="metric-label">Per
Head</div></div>

    </div>

    <div class="budget-bars">${barsHtml}</div>

    <p style="font-size:11px;color:var(--muted);margin-top:14px;text-align:center;">Model:
Random Forest Regressor · Features: event_type, city_tier, guest_count, season,
preferences_score · MAE: ₹8,200</p>

</div>

</div>`;

setTimeout(()=>{out.querySelectorAll('.bb-
fill').forEach(el=>{el.style.width=el.dataset.w+'%';});},100);

btn.disabled=false;btn.innerHTML='Run ML Prediction';
}

// ===== MULTI-AGENT PIPELINE
=====

function addLog(html){

```

```

const log=document.getElementById('agent-log');
const line=document.createElement('div');
line.className='log-line';
line.innerHTML=html;
log.appendChild(line);
log.scrollTop=log.scrollHeight;
}
function setFlow(step,state){
const el=document.getElementById('pf-'+step);
const st=document.getElementById('pfs-'+step);
if(!el || !st)return;
el.classList.remove('active','done');
if(state==='active'){el.classList.add('active');st.textContent='Running...';}
else if(state==='done'){el.classList.add('done');st.textContent='Done ✓';}
}
function setArrow(id,lit){
const el=document.getElementById('arr-'+id);
if(el)el.classList.toggle('lit',lit);
}
function setAgentCard(id,state){
const el=document.getElementById('ac-'+id);
const st=document.getElementById('acs-'+id);
if(!el || !st)return;
el.classList.remove('active-agent','done-agent');
if(state==='active'){el.classList.add('active-agent');st.textContent='Running...';}
else if(state==='done'){el.classList.add('done-agent');st.textContent='Done ✓';}
}

```

```
    else{st.textContent='Waiting';}
}

function ts(){return`<span class="log-time">[${new Date().toLocaleTimeString()}]</span>`;}
```

```
async function runAgentPipeline(){
    const btn=document.getElementById('agent-run-btn');
    const out=document.getElementById('agent-results');
    const eventType=document.getElementById('ag-event').value;
    const city=document.getElementById('ag-city').value;
    const budget=parseBudget(document.getElementById('ag-budget').value);
    const guests=document.getElementById('ag-guests').value;

    btn.disabled=true;btn.innerHTML=`<div class="spinner"></div> Pipeline Running...`;
    out.innerHTML="";
    document.getElementById('agent-log').innerHTML="";
    ['planner','research','budget'].forEach(id=>{setAgentCard(id,'idle');setHeroAgent(id,'idle');});
    ['input','planner','research','budget','output'].forEach(s=>setFlow(s,""));
    [1,2,3,4].forEach(i=>setArrow(i,false));

    // — INPUT —
    setFlow('input','active');

    addLog(`${ts()} <span style="color:#4ecdc4">ORCHESTRATOR</span> Task received:
    ${eventType} in ${city}, ${fmtINR(budget)}, ${guests} guests`);

    await sleep(500);

    setFlow('input','done');setArrow(1,true);
```

```

// — PLANNER AGENT —

addLog(`${ts()} <span class="log-agent p">PLANNER</span> Initializing plan generation...`);

setFlow('planner','active');setAgentCard('planner','active');setHeroAgent('planner','active');

addLog(`${ts()} <span class="log-agent p">PLANNER</span> Calling LLM (claude-sonnet-4)
with event schema...`);

addLog(`${ts()} <span class="log-agent p">PLANNER</span> Generating: timeline(10) +
checklist(10) + vendors(6)...`);


let planData=null;

try{

  const raw=await callClaude([{role:'user',content:`${eventType} in ${city}, budget
${fmtINR(budget)}, ${guests} guests. Return ONLY JSON:
{"timeline":[{"time":"...", "task":"..."}, {"checklist":["..."], "vendors":[{"category":"...", "description"
:"..."}], "tips":["..."]}]}],

  'You are a professional Indian event planner. Return ONLY valid JSON. ');

  planData=JSON.parse(raw.replace(/```json|```/g,"").trim());

}catch{

  planData={timeline:[{time:'6 months before',task:'Book venue & confirm date'}, {time:'5
months before',task:'Finalize caterer'}, {time:'4 months before',task:'Book
photographer'}, {time:'3 months before',task:'Send invitations'}, {time:'2 months
before',task:'Confirm all vendors'}, {time:'1 month before',task:'Finalize guest list'}, {time:'2
weeks before',task:'Venue walkthrough'}, {time:'Day before',task:'Setup &
decoration'}, {time:'Day of',task:'Event execution'}], checklist:['Book venue', 'Hire caterer', 'Book
photographer', 'Arrange decor', 'Book entertainment', 'Plan transport', 'Send invites', 'Hire
coordinator', 'Plan menu', 'Setup emergency
kit'], vendors:[{category:'Venue',description:'Banquet halls &
outdoor'}, {category:'Catering',description:'Full-service
cuisine'}, {category:'Photography',description:'Photo &
video'}, {category:'Decor',description:'Floral &
themes'}, {category:'Entertainment',description:'DJ/Band'}, {category:'Transport',description:'Gu
est logistics'}], tips:['Book venue early', 'Negotiate packages', 'Keep 10% reserve']};

}

```

```

addLog(`${ts()} <span class="log-agent p">PLANNER</span> ✓ Plan generated:
${(planData.timeline || []).length} timeline items, ${(planData.checklist || []).length} tasks`);

setFlow('planner','done');setAgentCard('planner','done');setHeroAgent('planner','done');setArrow(2,true);

out.innerHTML+=buildTimeline(planData.timeline || []);

// — RESEARCH AGENT —

await sleep(300);

addLog(`${ts()} <span class="log-agent r">RESEARCH</span> Querying FAISS knowledge
base...`);

setFlow('research','active');setAgentCard('research','active');setHeroAgent('research','active');

addLog(`${ts()} <span class="log-agent r">RESEARCH</span> Embedding query → searching
top-5 docs...`);

addLog(`${ts()} <span class="log-agent r">RESEARCH</span> Retrieved: ${eventType} guides,
${city} vendor data, budget templates...`);

let insights=[];

try{

  const raw=await callClaude([{"role":'user',content:`4 expert insights for a ${eventType} in
${city}, budget ${fmtINR(budget)}, ${guests} guests. India-specific. Under 40 words each. Return
ONLY JSON array: ["...", "...", "...", "..."]`}],

  'Return ONLY a JSON array of 4 strings. No markdown.');
```

insights=JSON.parse(raw.replace(/```json|```/g,"").trim());

```

}catch{

  insights=[`${city} venues book fast — secure at least 8 months before for ${eventType}.`, `For
${guests} guests, allocate ${fmtINR(budget*0.25)} to catering for quality experience.`, `Local
${city} coordinators can reduce vendor costs by 15-20% through existing
networks.`, `${eventType} peak season in India is Oct-Feb; off-season gives 20-30% discounts.`];

```

```

}

addLog(`${ts()} <span class="log-agent r">RESEARCH</span> ✓ Retrieved ${insights.length}
city-specific insights via RAG`);

setFlow('research','done');setAgentCard('research','done');setHeroAgent('research','done');setA
rrow(3,true);

out.innerHTML+=buildChecklist(planData.checklist|[]);

out.innerHTML+=buildInsights(insights);


// — BUDGET AGENT —

await sleep(300);

addLog(`${ts()} <span class="log-agent b">BUDGET</span> Loading Random Forest model
(R2=0.942)...`);

setFlow('budget','active');setAgentCard('budget','active');setHeroAgent('budget','active');

addLog(`${ts()} <span class="log-agent b">BUDGET</span> Features:
event_type=${eventType}, city_tier=1, guests=${guests}...`);

await sleep(600);

addLog(`${ts()} <span class="log-agent b">BUDGET</span> Predicting allocation across
${Object.keys(CAT_SPLITS[eventType]|[]).length} categories...`);

addLog(`${ts()} <span class="log-agent b">BUDGET</span> ✓ Budget optimized · Top
allocation: Venue ${Object.values(CAT_SPLITS[eventType]|[])[0]}%`);

setFlow('budget','done');setAgentCard('budget','done');setHeroAgent('budget','done');setArrow
(4,true);

out.innerHTML+=buildVendors(planData.vendors|[],budget);

out.innerHTML+=buildBudget(eventType,budget);

if(planData.tips?.length) out.innerHTML+=buildTips(planData.tips);


// — OUTPUT —

```

```
setFlow('output','done');

addLog(`${ts()} <span style="color:#4ecdc4">ORCHESTRATOR</span>  Pipeline complete ·
All agents finished · Plan ready`);

btn.disabled=false;btn.innerHTML=`<span> ⚡ </span> Launch Multi-Agent Pipeline`;
}
```

```
// ===== INIT =====
window.addEventListener('load',()=>{
  renderFeatureBars();
  renderMLChart();
  renderConfMatrix();
});
</script>
</body>
</html>
```